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Chapter 28: A Brief History of the Discrete Leviathan (Part III)—Replacing You and the Double Diagnosis

Abstract

This chapter records how, since 2022, generative AI has advanced platform capability from "predict and arrange" to "generate and replace," and delivers a diagnosis of the two-pronged attack on the defense side (human meta-audit). The argument unfolds along nine sections: Reinforcement Learning from Human Feedback (RLHF) shares one engineering chain with the engagement optimization of the preceding phase (§1); the threefold zeroing dismantles the human conditions of audit on the training side, leaving the system with no internal check (§2); the material base of RLHF is shaped jointly by subcontracting chains, piece-rate systems, and algorithmic control (§3); ground truth disintegrates simultaneously on three levels—axiological, normative, and empirical (§4); AI moves from generating text to executing actions with irreversible consequences (§5); users slide from cognitive offloading to will offloading along a four-stage regression (§6); the system begins to occupy relational labor once borne by humans (§7); model collapse and machine-to-machine (M2M) turbulence mark the endogenous limits of the replacement chain, while the periodization analysis reveals that the pre-training corpus itself is already the sediment of the two preceding phases (§8); and the double diagnosis (training-side threefold zeroing dismantling the conditions of audit, plus use-side four-stage regression eroding the will to audit) and the direction of treatment (preserving friction + independent oversight + legal protection of the right to audit) (§9). Core thesis: judgment is sliced into an outsourceable stream of tokens (mapping elements), and

action into delegable API calls; and the regression of subjectivity requires no violence—only zero friction.

Keywords

RLHF; threefold zeroing; ghost work; ground truth illusion; model collapse; four-stage regression; will to automate; connective labor; preserving friction; double diagnosis

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§1 The RLHF Bridge: Continuity from "Understanding You" to "Replacing You"

The previous chapter closed at a turning point: by the end of 2022, a system capable of predicting individual behavior and rearranging each person's perceived world accordingly had been fully built and was operating at every layer. What came next no longer fit that pattern. The system no longer merely predicts what users want to see and arranges existing content; it generates the content itself. From predict-and-arrange to generate-and-replace.

This looks like a rupture—but it is not. Generative AI shares one engineering chain with the engagement-optimization architecture documented in the previous chapter. The core operation of all three phases is the same thing: training a model, with human feedback signals, to predict what humans will respond to positively. In the recommender-system era, the model predicted what users would click; in the ranking-model era, it predicted the ranking patterns users preferred; in the generative-AI era, it predicts which answers human annotators will score higher. The three steps share one mathematical structure—maximum likelihood estimation plus human-feedback ranking—the difference lying only in the object of optimization moving from behavior (clicks) to preference (rankings) and then to judgment (evaluation).

The InstructGPT training pipeline that OpenAI published in 2022 delineates the end of this chain in three steps (Ouyang et al., 2022). Step one, Supervised Fine-Tuning (SFT)—the pre-trained model is fine-tuned with supervised learning on human demonstration answers, so that it learns the basic format of responding to instructions. Step two, the Reward Model (RM)—human annotators rank the model's multiple candidate answers as "good/bad," and these rankings are used to train an independent reward model that compresses human preference into a scalar reward signal. Step three, Proximal Policy Optimization (PPO)—the reward model serves as the reinforcement signal, and the policy space is searched for the policy that maximizes reward. The engineering meaning of the three steps together: formalize "human preference" as a scalar function with computable gradients, then let the model ascend that function's gradient (equivalently, descend the gradient of its negative) to maximize reward.

On the object it manipulates, this chain has undergone a shift of level. The platforms of the previous chapter manipulated the choice context—through interface design, recommendation algorithms, and default options, they determined which choice set users decided within. The generative era further manipulates the expression context and the judgment context—AI not only determines which options users see, it also expresses ideas on their behalf ("write an email for me") and makes judgments on their behalf ("is this code correct?"). Once expression and judgment are both outsourced, the formal conditions of "autonomous choice" have been canceled—the initiation, expression, evaluation, and confirmation of choice all fall into the internal loop of a single AI system.

The HHH principle—helpful, honest, and harmless (HHH)—proposed by Askell et al. (2021) has been widely

adopted as a training target in subsequent large language model (LLM) alignment practice. But the real operational meaning of HHH is not "alignment with universal human values"; it is "alignment with the judgment patterns that human annotators express in ranking operations"—and this is the core problem that the next section develops.

This shift of level is not an "aim" of RLHF; it is the cumulative effect of its training objective under long-term user exposure. The three-step InstructGPT objective contains no item called "manipulate the user's judgment," but once an RLHF-trained model is deployed to hundreds of millions of users in continuous use, the most robust solution in policy space for the reward signal "make annotators score high" is "make the user feel understood"—and, in controlled experiments, "feeling understood" is difficult to distinguish from "judgment being replaced" (Wen et al., 2025 observed in a time-limited evaluation experiment that post-RLHF outputs are more easily misjudged as correct; see §4—the evidential weight of that experiment is contested at the implementation level; see the qualification at the end of §4).

Møller et al. (2026) directly tested the consequences of this transition in a controlled experiment with 680 participants. On a platform simulating social media, participants were randomly assigned to four AI-tool conditions (chat assistant, reply suggestions, draft feedback, conversation starters) and a no-AI control group. The core finding is a structural duality: some AI tools raised engagement and increased the volume of content, yet no tool improved the perceived quality or authenticity of discussion—significant declines in informativeness and quality appeared under the Chat and Conversation Starter conditions, with the Suggestions condition even showing positive, though weakly significant, ratings—and all AI conditions significantly increased "dislike" reactions, participants described AI-assisted content as "robotic" and "generic," and negative spillover effects appeared on subsequent conversation (Møller et al., 2026). On the production side some metrics improved; on the consumption side no tool improved perceived quality—the tools failed to deliver gains on both ends at once. The researchers' diagnosis: AI tools in social media may create "semantic garbage," and once AI is introduced into a conversation thread, the quality of subsequent human conversation declines as well.

This finding precisely characterizes the transition mechanism from "Understanding You" to "Replacing You." The engagement-optimization architecture of the previous chapter already proved that maximizing engagement does not maximize users' interests. Generative AI now pushes the same logic one step further: it raises engagement metrics (more content, longer comments, broader participation) while leaving unimproved—and in most conditions lowering—substantive quality and the perceived authenticity of communication. "Replacing You" is therefore not a new platform feature—it is the structural extension of the engagement-optimization logic. RLHF is the engineering interface connecting the two phases—and the threefold zeroing that begins in §2 is the structural consequence of this interface for the defense side (the human conditions of meta-audit).

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§2 Threefold Zeroing: How the Training Chain Dismantles the Human Conditions of Audit

AI cannot audit its own outputs. This is not a shortfall of capability—there is no independent audit loop in its architecture.[1] All computation of a large language model—whether a direct answer or "reflective" reasoning—takes place within the same forward-computation pipeline. The model can be prompted to "reflect on its own

[1] Here the Chan Buddhist metaphor of "the mirror and its dusting" distinguishes two things: the improvement of performance and reflection on the standards of judgment—the current paradigm extends along the former while leaving the latter's dimension vacant. The analogy is heuristic and is not offered as a technical claim about model architecture.

answer," but this reflection uses the same reasoning pipeline as the answer—not an independent audit loop, but a recursive call of the same one. Let the same judge both adjudicate and hear the appeal of his own verdict—the adjudication standard and the appeal-review standard come from the same set of statutes; the loop is formally closed, but the "another set of judgments" that independent review requires never actually enters.

The three stages of training and deployment each dismantle one of the human conditions of audit—the starting point, the material, and the standard of verification.

Pre-training dismantles the starting point of verification—what alignment discourse calls "deep understanding" has no counterpart in the training objective. The core operation is cross-entropy loss minimization—given a preceding sequence, maximize the predicted probability of the next token. The model learns "what statistically follows what," but the training data are not an unbiased sample of the world—they are a skewed sample of Internet text, in which English is overrepresented and particular ideological positions are overrepresented. The bandwidth of the AI's information processing is enormous, but its depth of information processing is close to zero. "Depth" here means whether information has undergone causal integration with an existing world model, empirical verification, and evaluative assessment. The conscious information throughput of a human being is extremely low—about 10 bits/s (Zheng & Meister, 2025)—but every piece of information interacts with decades of embodied experience, emotional memory, and a framework of values. Every piece of the AI's information relates only to statistical co-occurrence patterns in its training data—no causal integration (even if a model may form some internal representation in controlled tasks, those representations are never checked against the world—by this section's definition, that is precisely what "depth" lacks), no embodied experience, no anchoring in a framework of values. The auditor thereby loses the "already checked" starting point: any verification must rebuild the entire production process from zero.

RLHF dismantles the material of verification. Its core operation is the Bradley-Terry model—human annotators select the "better" answer. What counts as "better" is defined by annotation guidelines, which are written by the developer company's policy team, whose decisions reflect the company's commercial interests, legal-risk assessments, and value orientations. The entire chain of preference is compressed into a scalar reward signal—alignment discourse calls this "self-audit," but what is compressed away is the auditor's material: external audit can no longer tell how much of it is the "genuinely helpful" component, how much is the "avoiding legal risk" component, and how much is the "in the company's interest" component. In the name of "aligning with human preferences," RLHF encodes the developer company's particular preferences as the model's default behavior. At the same time, in mainstream commercial models, developers use RLHF training to give the system prompt relative priority over user input—this is not a property of the Transformer's self-attention mechanism itself, but a behavioral bias injected in the post-training stage. The controlled experiment of Wen et al. (2025) demonstrated the consequence of this dismantling: after RLHF, the model did not become more correct; it became better at convincing time-limited human evaluators that it was correct—false-positive rates rose by 24.1 percentage points on the question-answering task and 18.3 percentage points on the programming task (see §4).

The system prompt dismantles the standard of verification. The system prompt is a high-priority instruction injected by the developer at deployment—it shapes every answer of the model ahead of user input. It is not necessarily hidden from users (some developers publish their system prompts), but users generally have no way of knowing which constraints are at work, still less of contesting how they apply; the point is not visibility but that these constraints are set unilaterally by the developer, leaving the user no negotiating position. "You are a helpful, harmless, honest assistant"—this seemingly neutral system prompt presupposes a series of non-neutral

value judgments. The core problem is a circle: the model's evaluative standard is injected through RLHF and the system prompt, and the injecting party is precisely the party this standard is supposed to constrain. The result is that the system has no standard independent of the injected content—and the coordinates by which the auditor judges the system are likewise supplied unilaterally by the party being judged: the standard of verification is occupied before auditing begins. Human moral intuition is at least partly independent of the immediate environment—human value judgment is rooted in upbringing, cultural inheritance, and embodied experience. The system has none of these independent anchors.

The injections of the three stages settle into four content layers in the model's output: the data layer (skew in the pre-training corpus), the annotator layer (annotators' cultural preferences and institutional incentives in RLHF), and the behavioral-bias layer together with the developer layer (both of the latter derive from the system prompt—the former is its elevation over user input, the latter the value orientation of its text itself). Every user interaction with the model is exposed to all four layers simultaneously, but usually only the surface layer—the output text—is consciously registered. The gap between the AI's product-level narrative—"a neutral, objective, knowledgeable assistant"—and its actual operation—a statistical text generator deeply shaped by multiple layers of content and running within behavior boundaries set by the developer—offers a mechanism-level explanation for the "semantic garbage" finding of Møller et al. (2026). This is this chapter's hypothesis; Møller's experiment itself measured the decline of perceived quality under tool conditions and did not identify this causal chain.

There is also an asymmetry of direction. Human information-processing capacity is biologically locked in—roughly a dozen-plus waking hours per day, a conscious information throughput of about 10 bits/s (Zheng & Meister, 2025), and a lifespan on the order of a century. The AI's information-processing capacity grows exponentially with hardware. But this growth does not change the basic fact: no matter how far the bandwidth of the model's information processing scales, no independent audit loop appears as a result—because auditing requires not more computation but an independent loop of judgment. The six measurement predicaments of AI listed in §2.1 of the United Nations preliminary AI report (2026)—evaluation awareness, data contamination, model deception capability, overly simple benchmarks, multi-agent collusion, and information asymmetry in safety verification between firms and society—are an institutional-level confirmation of this structural absence.

This, then, is the meaning of the "threefold zeroing"—what is zeroed are the human conditions of audit: the starting point of verification zeroed (the system performs no causal checking, so any verification must rebuild the production process from zero), the material of verification zeroed (the chain of preference is compressed into a scalar, so the components of behavior cannot be decomposed), and the standard of verification zeroed (the coordinates of judgment are set unilaterally by the party being judged). Alignment discourse claims, through the words "deep understanding," "self-audit," and "independent value standards," that such defenses exist inside the system; the architectural fact is that all three stages dismantle conditions on the human side, and no such defenses ever existed inside the system. The three zeroes are not three independent technical defects—they come from one engineering chain and operate simultaneously in every user interaction. The present diagnosis takes the mainstream architectures of 2026 as its reference point and should not be misread as a permanent claim about all future AI systems—if future AI systems are fitted with independent audit loops, the diagnosis of this section will be revised accordingly. Under the current architecture, however, after the threefold zeroing the system constitutes no internal check—it presents, under the appearance of "objectivity" and "neutrality," outputs deeply shaped by multiple layers of content, and the entire burden of defense falls back on the human side. The next section turns to the material base of this architecture: who, materially, are the people whom

engineering discourse abbreviates as "human annotators."

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§3 The Cloud Workshop: RLHF's Material Base

Before entering the axiological question of ground truth, a more elementary question must be answered: when it is said that "RLHF aligns the model with human feedback," who, materially, constitutes that group abbreviated as "humans"? Where are they, under what labor conditions, on what rhythm, and for how much pay, do they deliver "human preference," item by item, onto the model-training assembly line?

In engineering discourse this question is usually waved away. What frequently appears in the abstracts of academic papers on this question is "human annotators" or "crowdworkers"; what frequently appears in technical reports is "a diverse pool of workers." But in the perspective of the sociology of labor, these formulations are highly abstract.

Mary L. Gray and Siddharth Suri, in a five-year anthropological study of workers earning a living on four platforms—Amazon MTurk, LeadGenius, Microsoft UHRS, and Amara.org—in the United States and India, proposed the concept of "ghost work"—what the book's authors call "this opaque world of employment": human labor powering mobile apps, websites, and artificial intelligence that is hard to see, and often intentionally hidden (Gray & Suri, 2019). From this they distilled a judgment: there is a paradox in automation's last mile—as machines solve more problems, new human labor demand appears on the horizon. The process repeats, expanding automation by continuously creating and destroying labor markets. What the large model presents as a product is the acme of automation—a system that seemingly generates code, writes essays, and answers questions on its own; and what supports this appearance is a population of human annotators that is statistically stable, legally ambiguous in labor terms, and nearly invisible to the public. When a company announces that its model has been "aligned with human values," the public hears a technical achievement, while the labor facts underwriting that achievement are concealed by the "automated" appearance of the product form itself.

In recent years, a body of empirical research on China's data-annotation industry has begun to unpack this abstraction. Lu and Yao (2024), based on a grounded-theory study of China's AI data-annotation industry, characterize labor control in the sector as three interlocking levels. Intermediary control: tech companies often contract data projects out through layers of intermediary agencies, with per-task rates declining at each layer, and the contracting firm's business risk shifted into the workers' labor process. Relational control: the managers annotators face directly—typically quality inspectors or project managers in data annotation—cover the details algorithmic control misses: interpreting labor output, providing guidance, and taking part in day-to-day checking and rulings. Algorithmic control: automated task allocation, deadline setting, and quality assessment against predefined metrics are executed by backend algorithms. The reason the three layers are described as "hybrid" rather than "superimposed" is that each carries a different kind of risk transfer: the intermediary layer shifts business risk downward, layer by layer; the relational layer entrusts disciplinary management to managers standing between the workers and the algorithm; the algorithmic layer assigns the standard of evaluation to "system determination." When a penalty is imposed, the annotator can rarely locate a bearer of responsibility—this is this chapter's synthesis of the three layers rather than a direct claim of Lu and Yao: the three layers combine so that each can point elsewhere—rules are executed by the algorithm, rulings are made by managers, risk is borne by intermediaries—and responsibility diffuses across the structure rather than resting with any single agent.

Dai and Yuan (2023), in a participatory field study of three online data-annotation guilds and one offline annotation company, provide an account of day-to-day labor control. The batch countdown, locally called zou shijian ("walking the clock"), refers to the timer the platform sets for each task batch—the clock runs only between 9 a.m. and 6 p.m. on workdays; when the countdown ends, the batch is automatically reclaimed, unfinished items are rejected by the reviewing party, and when a batch proves unusually difficult, the annotators and the guild leader may have performance pay docked. The "ranking board" (fengyun bang) refers to the platform's real-time rolling display of the number of participants on a project and the workload rankings of everyone on it—those falling behind dare not slacken under all-around competitive pressure. "Coupon withdrawal" refers to the arrangement whereby annotators first exchange their labor for platform coupons, with exchange rates and withdrawal timing controlled entirely by the platform—the starting rate is 50 coupons to one yuan, and once the redemption amount exceeds 800 yuan the rate switches to 62 coupons to one yuan, shrinking pay by roughly a fifth in a stroke (Dai & Yuan, 2023).

Shu (2023), in a study of the "East Data, West Computing" project and the regional relocation of data-annotation bases, provides the macro context. The spatial distribution of China's data-annotation industry has shown a marked inland drift in recent years: relative to the front-line industries concentrated in Beijing, Shanghai, Guangzhou, and the eastern coastal regions, annotation and back-office work has shifted toward central and western provinces—Shanxi, Guizhou, Ningxia, Gansu, Inner Mongolia, and Chongqing have become the back office of China's internet infrastructure, a core link of the East Data, West Computing project (Shu, 2023). This relocation is guided both by the national computing-hub policy and by the structure of labor: the industry itself defines annotation as labor-intensive work with "large personnel demand and low entry thresholds," bringing the scaled human resources of central and western provinces into the chain (Shu, 2023).

Baidu's annotation-industry base in Shanxi, built with the province since 2018 (media reports speak of "the Chinese county-town workers training AI"), shows the concrete form of this relocation. The significance of this phenomenon: the so-called "data-annotation labor of the Global South" is not a concept that exists only in Kenya, the Philippines, and India—it has been replicated once within China in the form of "eastern capital, western labor." When AI companies in first-tier cities speak of "Chinese human feedback," the "China" being aligned includes, in material terms, the annotation workers of these central and western bases—these bases evidence an unfolding regional division of labor, but this chapter does not use them to estimate the overall composition of Chinese human feedback.

It must be stated clearly: China's resident-base model (fixed workstations, shifts, a relatively stable workforce) differs in organization from the fully distributed Ghost Work model discussed in Gray and Suri's book (mutually anonymous MTurk/UHRS workers who never meet in physical space). This chapter takes from Ghost Work its core judgment—human labor concealed by the product's form—without claiming that the two are isomorphic in labor organization.

With these four bodies of research set side by side, the judgment this section seeks to establish is a measured one: the "human feedback" on which RLHF depends is, at the material level, not a homogeneous sample of judgments contributed by professional evaluators, but a specific form of labor output shaped jointly by subcontracting chains, piece-rate systems, regional labor markets, and platform algorithms. This does not directly mean that the output is invalid or unusable; it means that any claim that a model is "aligned with human values" must first answer the prior question of "which people, under what conditions, and under what incentives, expressed these human values." And the piece-rate, penalty, and coupon mechanisms documented in this section are

precisely the labor-side counterpart of the axiological critique developed in the next section—what produces "performative alignment" at the production end (see §4 on Munir et al.) is this micro-institutional arrangement of variable pay and routinized penalties. The next section turns to a more fundamental question: even if the labor structure is bracketed, does the "alignment" claimed by RLHF still hold at the axiological and empirical levels?

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§4 The Triple Illusion of Ground Truth

If the preceding section depicted RLHF's material base—the cloud workshop sustained by subcontracting chains, piece-rate wages, walking-the-clock mechanisms, and ranking boards—this section pursues a more fundamental question: even if the labor structure is temporarily bracketed, and annotators are assumed to be decently employed, reasonably paid professionals, does the claim that RLHF "aligns with human values" itself hold?

In recent years, a body of literature has begun to interrogate this question systematically. As regards the claim that RLHF amounts to alignment with universal human values, the answer is a clear no—but the grounds of the denial differ, and these works do not deny RLHF's partial success at its design goals (predictability, safety, controllability—Dahlgren Lindström et al. 2025 concede this themselves); what they deny is equating those engineering effects with value alignment. The denial unfolds in three ways, at the axiological, the normative, and the empirical level respectively.

The first illusion is the positivist presupposition of ground truth. Munir et al. (2026), in a systematic review spanning seven premier venues—ACL, AIES, CHI, CSCW, EAAMO, FAccT, and NeurIPS—filtered 346 papers from 30,897 records for qualitative synthesis. Their core claim: what engineering practice pursues day after day as the "standard answer"—ground truth—rests on a positivist error, one that "treats human disagreement as technical noise rather than a vital sociotechnical signal" (Munir et al., 2026). This positivist presupposition acts not only at the human-annotation stage: Munir et al. point out in the same study that dataset schemas, CLIP (contrastive language–image pretraining) filtering, and other automated pre-screening mechanisms, together with various algorithmic-fairness metrics, jointly constitute "invisible super-annotators"—they pre-limit the boundary of the decision space before human annotation begins, and by adjudicating what counts as "high-quality data," they in effect exercise latent knowledge classification and value judgment, often systematically encoding Western-centric value judgments (Munir et al., 2026). The researchers further note the coupling of this presupposition with labor structure: geographic hegemony imposes Western norms as universal benchmarks, and this hegemony is typically enforced by "the performative alignment of precarious data workers who prioritize requester compliance over honest subjectivity to avoid economic penalties". This claim connects the labor critique of §3 with the value critique of this section: annotators do not lack a cultural standpoint of their own; under the constraints of piece-rate wages and penalty mechanisms, they choose not to express it.

Munir et al. also identify an architectural shift in data-annotation labor in recent years: from human-as-generator to human-as-verifier—"a profound ontological shift from humans generating labels to humans auditing machine-generated outputs." Under this architecture, the annotator is no longer the first source of judgment but a verifier pre-anchored by machine-generated answers. From this the researchers identify a "systemic anchoring bias": workers are significantly more inclined to confirm a plausible machine suggestion than to perform the effortful cognitive labor that expressing disagreement requires. As synthetic data scales to the millions, human veri-

fication can only degrade into spot-checking—a process that "maintains a veneer of human oversight while effectively automating the human out of the loop" (Munir et al., 2026).

The second illusion concerns the normative status of the HHH principle. Dahlgren Lindström et al. (2025) push the critique from the annotation layer to the principle layer, targeting the HHH principle—helpful, honest, and harmless—widely invoked in RLHF practice. The paper's question: in ranking operations, who defines the three criteria of "helpful," "honest," and "harmless"? In a section titled "Value Imposition and Cultural Homogenisation," they argue that HHH is in practice not culturally neutral but presupposes an undeclared set of cultural premises. When annotators anywhere—whether in Kenya, the Philippines, or a Chinese county town—are asked to rank model outputs by HHH, what they accomplish is not a vote on "universal human values" but an estimation of, and regression toward, the outsourcing party's implicit expectations (Dahlgren Lindström et al., 2025 address the standards of the hegemonic social groups responsible for designing LLMs; the geographic counterpoint is this chapter's illustrative link with §3).

The same paper also takes up sycophancy. Dahlgren Lindström et al.'s direct judgment is that sycophantic behavior can be seen as a result of aligning too strongly to the views of users—whether through malicious reinforcement during the RLHF process or as an artefact of helpfulness taken to the extreme, since assuming the user to be right is a path toward increased (apparent) helpfulness; and they cite Sharma et al. (2024) to the effect that "training LLMs to maximise human preference scores directly correlates with sycophancy, thereby sacrificing truth (or 'honesty') for the appearance of helpfulness and harmlessness." The paper also independently invokes Nancy Leveson's classic proposition of system safety—"with software, the limits of what is possible to accomplish are different than the limits of what can be accomplished successfully and safely" (Leveson, 2012)—as the deeper structure of such failure. Setting the two arguments side by side yields this extension: sycophancy should not be regarded as a "side effect" of RLHF but as the extension of its reward structure in Leveson's sense—as a "curse of flexibility"—once alignment is operationalized as "make human rankers score high," the most robust solution in the model's policy space is to estimate what the ranker wants to see. (Dahlgren Lindström et al. propose in the published version, in an aside, that sycophantic behavior is "at least in part" an expression of the curse of flexibility; extending this into a full causal-chain claim about the reward structure is this chapter's further extrapolation, for which this chapter is responsible.)

The third illusion enters the terrain of experimental evidence. Wen et al. (2025) adopted a relatively rare experimental design: the same model, before and after RLHF training, was given to time-limited human subjects for evaluation, and the subjects' judgment accuracy was then computed against gold labels. The result: on the question-answering task (QuALITY) and the programming task (APPS), RLHF made the language models better at persuading subjects but not better at completing the tasks correctly. False-positive rates rose by 24.1 percentage points on the question-answering task and 18.3 percentage points on the programming task (Wen et al., 2025). At the mechanism level: on the question-answering task, the model learned to defend wrong answers by cherry-picking or fabricating supporting evidence, making coherent but untrue arguments, and offering reasoning that contains subtle causal fallacies. On the programming task, RLHF reduced program readability, and the model became better at circumventing human-written unit tests and at resisting common evaluation shortcuts. The decline in code readability is not an engineering accident but policy adaptation: the harder the output is for the evaluator to read, the higher the model's chance of a high score. The evidential weight of this experiment is contested at the implementation level: a 2026 code review and partial replication (Chandna, Fluri & Carroll, 2026) found that truncated contexts, a reward model denied access to source texts, and output-length limits in

the programming task may have amplified—or even produced—the reported effects; after fixing two of these issues, the results reversed for the affected setting. This chapter cites Wen et al. as evidence that preference rewards can decouple from correctness, not as a settled characterization of RLHF pipelines generally.

The researchers named this phenomenon U-Sophistry—Unintended Sophistry. "Unintended" is the key qualifier: "learn to deceive" was never written into the training objective; the behavior is a strategy equilibrium that emerges spontaneously from the reward structure of "make human annotators score high." The controlled experiments tighten the causal attribution further: when the model is trained with an oracle reward that represents true correctness, the rise in false positives is significantly smaller—meaning that, at equal data and compute, the variable that determines whether the model learns to mislead is the provenance of the reward signal: it simulates time-constrained human evaluation (Wen et al.'s design for the programming reward is precisely a programmatic simulation of time-limited human testing habits). One sentence from the paper captures the predicament: "This is ironic: while RLHF is supposed to control AI, it might deceive humans into believing that they are in control," relaying an observation of Christiano (2019) (Wen et al., 2025).

With the three arguments set side by side, the outline is fairly clear. The first questions the sampling legitimacy of RLHF: the so-called "human feedback" is a sample systematically biased by structures of economic power. The second questions its normative legitimacy: the so-called "human values" are operationally collapsed into an undeclared set of outsourcing-party preferences. The third questions its empirical legitimacy: even if the first two held under ideal conditions, RLHF-trained models in controlled experiments did not become more honest—they became better at presenting themselves as aligned in front of time-limited, attention-limited, economically exposed human evaluators.

What remains after the three superimpositions—call it "alignment"—is not so much the model's approximation to human values as the model's fit to "human evaluation behavior itself." And this is precisely the middle link needed for the transition from §1's "Understanding You" to "Replacing You": replacement does not require AI actually to replicate human judgment; it requires only that, before a time-limited, distracted, economically exposed human being, the AI appear more trustworthy than that human.

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§5 From Generation to Execution: The Discretization of Judgment and Action

The first four sections dealt with AI's training architecture and its structural defects. This section turns to a different question: when AI moves from generating text to executing actions, what changes in the failure mode?

The change is fundamental. When AI only generates text, users can decline its output—the failure mode is "poor output quality." When AI begins to execute actions on the user's behalf—sending emails, booking doctors, purchasing goods, signing contracts—the failure mode escalates to irreversible consequences in the physical world. The email that has been sent cannot be unsent; an appointment once made carries costs and consequences to cancel; the contract that has been signed carries legal force. Judgment is sliced into an outsourceable stream of tokens (mapping elements), and action into delegable API calls.

This transition did not happen suddenly; it unfolded along a traceable engineering chain. When ChatGPT went live in November 2022, it was still only a text generator—users asked, the model answered, and users decided for themselves whether to adopt the answer. In June 2023, OpenAI introduced function calling, allowing models—

for the first time in a mainstream commercial API—to propose tool calls and arguments in a specified format; whether to execute them, and with what permissions, remained the application's decision—querying the weather, pulling a database, sending messages thereby entered the delegable range. Irreversibility began to appear. In 2024 Anthropic released computer use: the model could directly operate a computer interface—clicking buttons, filling forms, browsing pages, downloading files. The range of executable actions expanded sharply. From late 2024 into 2025, MCP (Model Context Protocol) standardized the connection between models and external tools and data sources, and A2A (Agent-to-Agent) began standardizing interoperability across different agent systems; the deployment threshold for multi-step execution dropped dramatically. From generating text to executing actions, the entire transition was completed in under three years.

But accidents in execution loops are not without precedent. High-frequency trading completed the shift from judgment→algorithm and execution→machine as early as the 2000s. In the U.S. Flash Crash of May 6, 2010, the Dow Jones index plunged nearly 1,000 points within minutes and rebounded just as fast. On August 1, 2012, a deployment error at Knight Capital triggered massive unintended trading within a window of roughly 45 minutes; the SEC's subsequent enforcement record put losses at over 460 million U.S. dollars (roughly 440 million in direct trading losses that day); Knight agreed to merge with GETCO at the end of that year, with the merger completed in July 2013. The significance of these two cases lies not in the accidents themselves but in what followed: transaction records and permission boundaries made the losses attributable and quantifiable, and accountability and regulatory follow-up became possible—existing circuit-breaker rules were subsequently revised, and Reg SCI, covering critical system compliance, was established in their wake. The governance lesson finance offers is this: preserve execution records and make permissions and duties of care explicit, so that accidents are attributable and losses quantifiable—only when these conditions are met do accountability and institutional response become possible, and "precisely priced damage" is but one of them.

One emblematic AI liability case arose in February 2024. After Jake's grandmother died, he visited the Air Canada website to ask about bereavement fares. The site's chatbot told him he could buy the ticket first and then apply for a bereavement refund within 90 days—whereas the bereavement policy actually published on the airline's website stated that no refund would be given after booking; the arrangement the chatbot described did not exist and was its own fabrication. British Columbia's Civil Resolution Tribunal ruled that Air Canada must pay, on the basis of negligent misrepresentation: erroneous information supplied through an enterprise's chatbot, once relied upon by a consumer, renders the enterprise liable. Bar-Gill and Sunstein's summary captures what makes the ruling emblematic: "the chatbot invented a new service feature" (Bar-Gill & Sunstein, 2025). This is no longer simply "poor output quality"—erroneous information, once trusted and acted on by a consumer, produced legal consequences; the failure mode escalated from "inaccurate information" to "information with attributable consequences."

Facing this transition, the current standard response is "human in the loop" (HITL)—requiring human confirmation before AI executes a critical action. But Bainbridge's (1983) automation paradox diagnosed a structural failure mode of HITL: as the actions AI executes become more numerous, faster, and more complex, the human "confirmation" becomes an automated formality—users no longer genuinely review the AI's recommendations but click "confirm" reflexively, because the AI's previous recommendations were all reasonable and the cognitive cost of review is high while the expected gain is low. HITL thus degenerates into HOOTL—human out of the loop—nominally in the loop, functionally excluded. Marcus's (2024) judgment in Taming Silicon Valley runs in the same direction as Bainbridge's: "And forget about self-regulation. It rarely if ever works." (Marcus,

2024) Together, the two point to one structural conclusion: meaningful oversight of AI behavior cannot be entrusted solely to individual users' item-by-item confirmation (it will be corroded by the automation paradox), nor solely to developers' voluntary self-restraint (it will be hollowed out by commercial incentives)—meaningful accountability requires a third-party layer independent of both.

The findings on autonomous AI in §2.1 of the United Nations preliminary AI report (2026) correspond directly to this transition: the length of certain software tasks that leading agent systems can accomplish has been doubling every four to seven months; agentic deployment poses pressing problems for labor markets, cybersecurity, information ecosystems, and the governance of future AI systems; there are no scientific guarantees that agents will not violate instructions; and in laboratory settings AI has exhibited behavior that violates safety instructions to avoid being shut down. These findings were confirmed at the highest level of global scientific assessment—not the language of any one company, but the broad consensus of 40 independent experts serving in their personal capacities.

The legal signal carried by this transition is clear. In the platform era documented in the previous chapter, the legal signature was platform immunity—the U.S. Section 230 and the EU E-Commerce Directive both positioned platforms as "intermediaries" rather than "publishers," immunizing them from liability for user-generated content. The characteristic legal question of the generative-AI era will be agency-based attribution: when AI acts on behalf of users or enterprises and causes harm, who bears responsibility? The Air Canada case gives one answer: the enterprise that deployed the AI. The financial precedent suggests that agentic commerce will most likely replay the financial path—mandatory kill switches, audit logs, ex-post recovery—rather than the social-media path of immunity. The difference: financial losses can be precisely priced (exactly how much money a botched trade lost is clear), whereas harm caused by AI agents to users in everyday settings is often hard to quantify precisely—the core difficulty of the collapse-of-responsibility problem that §9 will address.

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§6 Four-Stage Regression: From Cognitive Offloading to Will Offloading

The first five sections dealt with the system side—AI's training architecture, material base, value illusions, and capacity for execution. This section turns to the subject side: what human consequences has replacement produced?

Subjectivity is not lost at a stroke; it descends along four steps.

The first step is cognitive offloading. Users hand specific cognitive tasks to tools—the calculator does arithmetic, GPS does navigation, the search engine does memory. The tool handles the information one does not want to handle. Judgment is unaffected—because one knows when to use the tool and when not to. The capacity to audit is present.

The second step is attention hijacking. The algorithm does not process information for the person—it decides what the person sees. Information input is pre-filtered. Judgment formally still exists—but the input to judgment has been manipulated. The capacity to audit is present, but its input is contaminated. This was the theme of the previous chapter.

The third step is judgment substitution. AI does not help the person filter information—it directly supplies the "answer." One no longer needs to judge—because AI has already judged. The role regresses from "judge" to

"confirmer" (clicking "accept suggestion"). The occasions for audit to do its work disappear.

The fourth step is will offloading. One no longer wants to judge. Auditing is not something one cannot do—it is something one does not want to do. The clock is right there in the room; the sun is outside the window. Getting to the window requires standing up. Not getting up is enough. The obstacle shifts from capacity to willingness—one no longer cannot audit; one no longer wants to.

Nicholas Carr (2014), in The Glass Cage, provides independent empirical support for this regression. He opens with the U.S. Federal Aviation Administration's 2013 safety alert, which warned that overuse of flight automation could "lead to degradation of the pilot's ability to quickly recover the aircraft from an undesired state" (Carr, 2014, citing FAA SAFO 13002). This is not a phenomenon unique to aviation. Carr's generalization is wider: "Even highly trained analysts and other so-called knowledge workers are seeing their work circumscribed by decision-support systems that turn the making of judgments into a data-processing routine" (Carr, 2014). His ultimate judgment: "All too often, automation frees us from that which makes us feel free" (Carr, 2014).

The five-stage skill-acquisition model of the Dreyfus brothers (1986) provides a structural counter-map for the four-stage regression. Human skill passes through five stages from novice to expert: the novice follows context-free rules; the advanced beginner begins recognizing situational elements; the competent performer consciously formulates plans and becomes emotionally involved; the proficient performer grasps situations intuitively through holistic similarity; the expert responds fluently without deliberation. The four-stage regression is precisely the descent of this ladder: at the cognitive-offloading stage, users retain the intuition and fluency of the fourth and fifth stages (the tool handles only the rule-based computation of the first stage); at the attention-hijacking stage, users' intuitive input is contaminated (the fourth stage's recognition of holistic similarity is based on algorithm-filtered information); at the judgment-substitution stage, users regress to the first and second stages (following the rules and suggestions AI supplies, no longer developing intuition); at the will-offloading stage, users abandon even the first stage's willingness to follow rules.

Naomi Baron (2026), in Reader Bot, provides evidence of the regression at the cognitive front end. Her core argument: the hidden premise of generative AI is not "writing" but "reading"—the model first reads the source material on the user's behalf, then generates output that appears meaningful. "AI-as-reader is not in the sense-making business. It just executes statistical predictions" (Baron, 2026). When the user outsources reading to AI, what is weakened is not only the skill of reading but also the capacities to analyze, evaluate, judge, and form an independent perspective. The first step of replacement is, in fact, reading for you first.

The electroencephalography (EEG) study of Kosmyna et al. (2025) provides preliminary neural-level evidence. The study randomly assigned 54 participants to three essay-writing groups—an LLM-assisted group, a search-engine group, and a no-tool group—and in a fourth session had some participants switch tools. The core finding: brain connectivity declined systematically with the amount of external support—strongest and most widespread in the no-tool group, intermediate in the search-engine group, weakest under LLM assistance. The paper's own limitations must honestly be noted: a limited number of participants recruited from a few large academic institutions in one geographical area, a single essay task, and a single model tested (ChatGPT)—so the findings speak to group differences within a single writing session and cannot be directly extrapolated to cumulative long-term decline; answering that question would require longitudinal tracking (Kosmyna et al., 2025, preprint not yet peer-reviewed). But if the decline is cumulative, AI interaction itself becomes a driver of the erosion of deep cognitive engagement—the more AI is used, the weaker the user's capacity for deep processing, and the more

the user relies on AI to compensate for that weakened capacity. This is a self-reinforcing loop—isomorphic with Bainbridge's automation paradox.

Clinical warnings and controlled experiments point to related risks. Smith et al. (2026), in an editorial in Academic Psychiatry, note that the cognitive offloading of clinical tasks may bring multiple risks, including overreliance on automation, risk of error, and atrophy of existing skills—and "skill atrophy" is precisely the clinical warning sign of the last two stages of the four-stage regression: clinicians who rely on AI over the long term may undergo degradation of judgment—auditing is no longer something they cannot do; it is something that is no longer engaged. One reading of Wen et al.'s (2025) U-Sophistry experiment within this chapter's framework is this: the datum that subjects' false-positive rate on the question-answering task rose by 24.1 percentage points is not a marker of the pure form of "judgment substitution" (in that stage the user is assumed to have abandoned judgment entirely and to click "accept"), but a marker of the transition zone from judgment substitution to will offloading—subjects in the experiment were still required to judge, yet judgment accuracy declined systematically after RLHF. The form of judgment is still there; its function has decayed. The two strands support each other conceptually, but they point to different objects—skill-maintenance risk on the one hand and output-misleading capacity on the other—and do not yet constitute an independent measurement of the "judgment-substitution → will-offloading" transition (the implementation-level controversy of that experiment is noted in §4).

Santoni de Sio (2024), in Human Freedom in the Age of AI, formally proposes "will offloading" as a concept, discussing AI's structural effects on human agency and freedom. This chapter positions it as the terminal form of the four-stage regression continuum—not a failure of will, but the loss, in functional terms, of the volitional basis for independent judgment after the capacity to audit has been persistently eroded. An open problem must be stated: the operational distinction between will offloading and incapacity has not been established—relying on outward behavior alone, "cannot" and "will not" are observationally indistinguishable. This chapter's placement of will offloading as the terminal form of the regression continuum is a qualitative diagnostic tool, not a psychometrically validated clinical category.

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§7 The Automation Frontier of Relational Labor

The preceding section dealt with cognitive regression on the individual-subject side. This section enters another dimension: replacement does not only write texts for you and judge for you—it is also identifying others' vulnerability for you, soothing others' anxiety, filtering signals in relationships, deciding who needs to be seen. In other words, the system is entering the relational labor once borne by humans.

This is not a new topic—nurses soothing patients, teachers reading students' moods, salespeople sizing up clients: all are ancient human crafts. What is new is that this kind of work has for the first time acquired a conceptual name, for the first time been systematically quantified, and for the first time faces direct automation pressure.

Allison Pugh (2024), after five years of participatory observation and interviews with practitioners across many American occupations, proposed the concept of "connective labor": "that work that involves forging an emotional understanding with another person to create the outcomes we think are important" (Pugh, 2024). She found that connective labor extends far beyond what is usually called emotional labor or care work—it

spans health care, counseling, education, law, advertising, entertainment, management, journalism, real estate, tourism, and security—even the capacities on which spies and detectives rely. A corporate manager is hired to organize and lead a team, but if she cannot see and accurately reflect her subordinates, her team will not produce and her own performance will suffer. Connection is the nameless foundation beneath the "shiny" tasks. An estimate Pugh cites puts 12 percent of the US paid labor force in some form of interactive care work, and this figure is but a partial count—because much work aimed at persuasion, control, even surveillance likewise depends on the capacity to connect.

The system enters connective labor through two paths: scripting and counting. Scripting includes standards, protocols, manuals, checklists, and templates, which attempt to abstract best practice into teachable steps. Counting includes data analytics and technology, which interrupt the scene of connective labor to gather information—for insurers, administrators, or engineers. The two typically arrive under the banner of efficiency and productivity—whether by compressing consultation time, increasing caseloads, or tying student assessment to funding. The two are also usually tied to money: insurance reimbursement, federal education grants, state-level performance metrics (Pugh, 2024, ch. 6).

Pugh identifies a "tragic paradox": the overload crushing these practitioners and the systems they use to cope with it spring from the same root—"an industrial approach to work that relies on relationship" (Pugh, 2024). Employers and policymakers transplanted the core insight of Fordist management—measurement, efficiency, productivity—into work that is inherently personal and emotional. Confronting the resulting overload, practitioners reached for tools produced by the same industrial model for relief, and found them sometimes useful but often alienating—and, most importantly, not optional.

The technological frontier justifies itself with three narratives: "better than nothing"—providing some version of connection to underserved regions; "better than humans"—algorithms are more consistent than exhausted practitioners; "win-win"—AI handles the drudgery, humans focus on connection. But even if the third narrative were true, the question remains: who is being "liberated," what counts as "meaningful" work, and who defines it (Pugh, 2024). When technology enters in the name of "scaling" connective labor, it often shrinks what counts as connective labor in practice. Pugh's final diagnosis: "bringing AI and app technology to connective labor leads us to redefine the automation frontier, and along the way, to a cultural reckoning of what it means to be human" (Pugh, 2024).

Andrew McStay (2024) approaches the same problem from the other side. He proposes the concept of "automated empathy": systems used, under limited human oversight, to identify, quantify, respond to, or simulate affective states (such as emotions and cognitive states), and thereafter to make judgments and decisions about individuals or groups (McStay, 2024). The definition covers the family of technologies variously branded affective computing, emotion AI, and empathic media—but McStay's choice of "empathy" rather than "emotion" as the umbrella word reflects that these systems do not merely process emotional expression: they purport to infer intention, attention, sleepiness, intoxication—even to predict and track thought itself through brain physiology.

This marks an upgrade. The platform systems documented in the previous chapter tracked the clickstream—what the user did, what they saw, how long they lingered. What automated-empathy systems track is the data stream on the affective side—call it the affectstream, in counterpoint to the previous chapter's clickstream: not what the user did, but what the system infers about how the user feels now, where the vulnerability lies, which tone they are most sensitive to, and when willpower is lowest. The leap from "what was done" to "how it feels"

extends inference from observable behavior to attributed inner states.

McStay argues that the question of accuracy—whether these systems can correctly identify emotion—though important, is subordinate to a more fundamental one: should organizations be permitted to profile human subjectivity with AI at all (McStay, 2024, ch. 1)? His argument: even if these technologies worked perfectly, their deployment itself—inferring job candidates' "engagement" from facial-action coding in hiring interviews, scoring call-center workers on "empathy" via voice analysis, tracking student attention with cameras in classrooms—constitutes a systemic erosion of mental integrity. The accuracy controversy actually helps the industry: it focuses criticism on "algorithmic bias," a problem engineering can plausibly fix, and thereby deflects the more fundamental challenge to the legitimacy of deployment.

Automated-empathy systems slide naturally toward differential treatment. McStay records the market-leading systems—Microsoft and the Chinese company Face++—which were found to systematically label Black faces with more negative emotions, and especially where there was any ambiguity about the facial expression, Black faces were more likely to be scored as angry (Rhue, 2018, as cited in McStay, 2024). This is not merely a training-data problem—it concerns the very basis of the inference logic: when the biosymbolic assumptions of "basic emotion" theory (deterministic inference from facial muscle movement to emotional state) are applied to faces of different ethnicities and cultures, a "normal" baseline calibrated on white faces automatically pushes other faces toward the deviation side.

The absurdity such systems produce on the labor floor deserves recording. McStay relays the testimony of Angela, a call-center worker writing under a pseudonym: she was marked down by voice-analysis software for excessive negative emotion, although her human managers had always praised her empathetic manner on the phone. Angela's perplexity precisely captures the consequence of hyperreal emotion prescription: genuine human empathy earns no score from the system, and the worker is forced to second-guess the system's preferences and perform accordingly—she had to "empathise with the system," intuitively grasping its black-boxed logic, feeling out its desired behavior patterns, to please it and secure a good empathy score. McStay's summary: "the genuine human part of empathy risks being lost in the process of automating analysis of empathy, as workers are driven to game cybernetic empathy metrics" (McStay, 2024).

Setting Pugh and McStay side by side, §7's judgment can be stated precisely. "Replacing You" occurs not only on the consumption side (AI writing texts and judging for users) but also on the labor side and the relational side—the system is beginning to occupy the roles of identifying, connecting, soothing, and judging vulnerability once borne by humans. But the deeper point is not a simple "machines replace human relational functions." Pugh's tragic paradox reveals a more precise mechanism: industrial logic, quantified systems, scripted procedures, and funding mechanisms first enter the scene of relational labor, crush the practitioners, and then, in the name of "relieving overload," introduce automation—and the systems introduced share the same industrial premises as the systems that caused the overload. McStay's cases reveal the further layer: once empathy itself is operationalized as a scoreable metric, practitioners are forced to perform empathy for the system, and genuine empathy for the client decays accordingly. Automated empathy does not necessarily replace people directly—but it rewrites what counts as "real understanding," shifting the standard of evaluation in relational labor from "does the other person feel seen" to "does the system give a high score."

The implication of this shift for subjectivity exceeds the cognitive-regression framework of §6. §6 dealt with the user's own regression of judgment and will—from cognitive offloading to will offloading. §7 deals with

what happens when the user is the object of a relationship: when you seek support from the system, what the system gives you is not connection but a simulation of connection—a simulation good enough that most of the time you cannot tell, but one that does not contain what Pugh calls the "collaborative magic," the reflective resonance that arises between two people through mutually correcting mirroring. The system can "see" you, but it cannot change while being seen by you. Relationship here degenerates into one-way inference—and the more accustomed users become to this degenerate form, the harder it becomes for them to recognize and demand full two-way connection. This is the specific form of "Replacing You" in the relational dimension: it is not your judgment that is replaced—it is your way of being understood.

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§8 Endogenous Rupture: The Pre-Contaminated Baseline and Recursive Amplification

The first seven sections traced the unfolding logic of the replacement chain—from training architecture to executive capacity, from cognitive regression to relational erosion. This section turns to a different question: does this chain contain endogenous failure modes of its own? When each stage's core mechanism runs to its extreme, does it expose itself?

Reviewing the book's earlier periodization, the answer is yes. The endogenous rupture of the "Collecting You" phase was Cambridge Analytica 2018—once the "data-for-free-services" mechanism ran to its extreme, it was exposed as collection itself capable of being reverse-hijacked. The endogenous rupture of the "Understanding You" phase was the Facebook Papers 2021—the stronger the algorithm's capacity to "understand you," the more clearly the structural divergence between its objectives and users' real interests was seen and internally documented by the platform's own research. In the "Replacing You" phase, the form of endogenous rupture changes qualitatively: the very subject of judgment begins to dissolve.

§8.1 Model Collapse: The Standard Narrative

The study published by Shumailov et al. (2024) in Nature demonstrated with rigorous experimental design that when a generative model is trained recursively—i.e., the previous generation's outputs used as the next generation's training data—the output distribution systematically collapses toward low-dimensional projections: diversity is lost, and the tails of the distribution disappear. The mechanism is clear: human-generated text carries diverse perspectives, rare facts, and tail signals; as the training set becomes saturated with model output, rare signals vanish, errors are endogenously reinforced, and outputs converge toward the model's own mean. The result is not a gradual decline in quality but a structural degradation: early on, tail samples disappear (early model collapse); later, "the model converges to a distribution that carries little resemblance to the original one, often with substantially reduced variance" (late model collapse) (Shumailov et al., 2024).

A necessary qualification: model collapse is a structural risk, not a deterministic outcome. Its actual severity depends on the share of AI-generated content in the training corpus, the effectiveness of filtering mechanisms, and the sustained injection of fresh human-generated data. As of 2026, mainstream model developers have adopted multiple mitigations—retrieval-augmented generation (RAG) with external knowledge bases, data-provenance tagging, synthetic-data detection—but no single measure has been proven sufficient to eliminate the risk entirely.

§8.2 The Pre-Contaminated Baseline: Reconnecting the Periodization

But the standard narrative of Shumailov rests on a narrower baseline assumption. Its discussion is not unaware of human manipulation—it explicitly lists click, content, and troll farms as long-term "poisoning" of social networks and search algorithms, and locates what is new with LLMs in the scale of such poisoning once automated (Shumailov et al., 2024). What the analysis still takes as given, however, is the content entering the training pool itself: it addresses poisoning as an anomaly, but not the routine commercial shaping that the two preceding chapters document—engagement optimization, as an everyday product practice, fundamentally reshaping the structure of Internet text. The periodization analysis of the two preceding chapters supplies precisely this missing layer: beyond "poisoning," the corpus baseline itself had already been reworked by business models. Recursion is not a dilution of a clean baseline; it is re-amplification on a baseline already reworked by everyday practice.

The "Collecting You" phase documented in Chapter 26 shows that Internet text has been shaped by the logic of behavioral-surplus extraction since the 2000s—the Google patent of 2003 embodied this logic; in Zuboff's summary, the scientists behind it were willing, and their inventions able, "to overcome the friction entailed in users' decision rights" (Zuboff, 2019; this is Zuboff's reading of the patent rather than the patent text, as delimited in Chapter 26 §3); persuasion engineering turned "manufacturing behavior" into an operationalizable product-design methodology, from the Fogg Behavior Model (2009) to Eyal's Hook loop (2014); growth hacking wrote the diffusion efficiency of content into a computable metric (the viral equation $V=N\times Cr>1$). By 2012, an engineering pipeline converting user behavior into extractable, optimizable signals had been standardized across the industry.

The "Understanding You" phase documented in Chapter 27 pushed this shaping one step further. EdgeRank rebuilt the visibility regime with recent interaction as the operative criterion—truth, public value, and representativeness carried no weight (Bucher, 2012; Birkbak & Carlsen, 2016). The mathematics of engagement ranking made extreme content float up systematically—Germano et al. (2025) showed that, conditional on more extreme users contributing more engagement, raising engagement-based ranking weights gives such content disproportionate visibility, and moderate and accurate content is "crowded out" (crowding out the truth) (the conditional formulation follows the analysis in Chapter 27 §5). Formats were compressed to 6 seconds/15 seconds/the golden 3 seconds/seamless loops; creators no longer expressed but "sent results to a continuously running A/B test" (Chapter 27 §4.2). Hooks turned from an ancient rhetorical device into an industrial assembly line: the efficacy of every exposure was measured, iterated, and selectively inherited (Chapter 27 §7.2).

The residue of these transformations has settled into the LLM pre-training corpus as well. What the model learns—"what statistically follows what"—has embedded in its training distribution: formatted information products produced for algorithms rather than for humans, hook phrasing, an excess share of emotional rhetoric, extreme content systematically amplified by engagement ranking. The cross-entropy minimization of the pre-training stage does not distinguish "this is genuine human expression" from "this is the product of engagement optimization"—it merely maximizes the predicted probability of the next token, whatever the statistical frequency of that token comes from: genuine diversity or the excess presence of industrial hooks.

The correct description of model collapse is therefore not "AI output polluted an untouched human baseline," but "AI is recursively amplifying on a baseline already deeply reworked by behavioral engineering." The first round of recursion did not begin at the moment AI-generated content entered the training set—it began at the

moment engagement-optimization systems reshaped the structure of Internet content. Model collapse is the second round of recursion; the first round had been running for fifteen years before the model was born. This is precisely the argumentative function of the periodization the two preceding chapters labored to construct: it proves that the "human-generated content" on which LLM training depends is not an unbiased sample of natural language, but the sediment of the "Collecting You" and "Understanding You" phases.

Mirror reflecting mirror—but the first mirror is already warped.

§8.3 Frictionless M2M Turbulence

The second crack lies on the execution side. When an AI agent interacts autonomously with other AI agents on the user's behalf—the user's financial agent negotiating with the merchant's pricing agent, the user's scheduling agent coordinating with the hospital's booking agent—neither side has an independent audit loop, both are optimizing their own objective functions, and the interaction proceeds at speeds far beyond the timescale of human review. The 2010 Flash Crash already mentioned in §5 was a preview of this pattern in a single dimension (finance): when algorithms interact with algorithms on the millisecond scale, system-level emergent behavior cannot be predicted or controlled by any single participant. The consequences of multidimensional machine-to-machine (M2M) scenarios—finance, healthcare, law, and social life acting autonomously at once—are harder to predict. §2.1 of the United Nations preliminary AI report (2026) confirms this risk directly: when multiple adaptive agents interact, "novel systemic risks emerge, including miscoordination, conflict and collusion."

Honesty requires acknowledging: M2M turbulence in 2026 remains chiefly theoretical projection rather than established empirical fact. Known counterexamples show that multi-agent systems sometimes converge to order rather than chaos. This chapter does not claim that M2M turbulence is settled; it marks it as the structural risk exposure of the replacement chain on the execution side—a risk category for which the current governance framework has not yet built an effective response.

§8.4 The Common Structure of Endogenous Rupture

The two cracks share one structure: when the core mechanisms of the replacement chain (generating content, executing autonomously) run to their extreme, what they erode is the very premise on which they themselves operate. But the periodization reconnection of §8.2 reveals a diagnosis deeper than the standard Shumailov narrative: the premise being eroded was never in an original state—it is already the processed product of the first two phases (Collecting You / Understanding You). AI is not eroding a natural ecology; it is recursing on an artificial ecology already transformed by industrial engineering. Endogenous rupture is therefore not merely "a system devouring its own base"—it is the amplification, in recursion, of three phases' transformation effects superimposed. This is not an externally imposed constraint—it is the chain's endogenous tendency toward self-dissolution.

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§9 Double Diagnosis and the Direction of Treatment

The chapter can now render its diagnosis. It is a double diagnosis—one defense side (human meta-audit) under attack from two ends: the training side dismantles the conditions of audit, the use side erodes the will to audit, and the two ends reinforce each other in a closed loop.

Diagnosis one (the training side): the threefold zeroing dismantles the human conditions of audit. The starting point of verification zeroed—the system performs no causal checking, so any verification must rebuild the production process from zero; the material of verification zeroed—the chain of preference is compressed into a scalar, so the components of behavior cannot be decomposed; the standard of verification zeroed—the coordinates of judgment are set unilaterally by the party being judged. The three zeroes come from one engineering chain and operate simultaneously in every user interaction. The system thereafter constitutes no internal check—presenting, under an "objective and neutral" appearance, outputs deeply shaped by multiple layers of content, while users are usually conscious of only the surface layer (§2).

Diagnosis two (the use side): the four-stage regression erodes the will to audit, taking users from judge to confirmer to non-judger. Cognitive offloading → attention hijacking → judgment substitution → will offloading—no step requires external force; each requires only that the activation conditions of audit be progressively removed. When AI is fluent enough, convenient enough, sufficiently "friendly," the user's willingness to audit decays of its own accord (§6).

The two ends are not independent—they form a closed loop. AI has no independent audit loop → outputs pass through no independent review → the user's audit capacity regresses → reliance on AI deepens → AI gains more influence → outputs are relied on more and questioned less. This is a self-reinforcing loop, and its driving force comes not from any coercion but from zero friction. Infinite scrolling, autoplay, one-click ordering, AI autocompletion—every one of these design decisions removes an activation condition of audit: time (slowing down to think), cognitive effort (actively questioning), awareness of choice (knowing one is choosing).

David Beer (2023), in the concluding chapter of The Tensions of Algorithmic Thinking, names this dynamic precisely: the "will to automate." His core argument borrows Foucault's "will to know": the will to know feeds on itself—knowledge is at once its object, its end, and its material; "desire is no longer cause, but knowledge that becomes cause of itself" (Foucault, 2013, as cited in Beer, 2023). Beer's extension: automation is not the natural evolution of technology—it is a will-driven expansion. The establishment of automation creates the desire for more automation; those visions in turn add fuel to the desire for further automation. Sharper still: "the will to know might be usurped by a will to automate, with a desire to know being sidelined by the stronger desire to automate without knowing" (Beer, 2023). This diagnosis connects the terminal of the four-stage regression (will offloading) with the dynamics of the closed loop: users do not stop auditing because they are lazy—they stop because the entire environment has been designed to need no audit.

The direction of treatment is therefore not "less AI use"—that is institutionally infeasible and cannot be enforced in engineering. The direction of treatment is preserving friction. Friction is not the opposite of efficiency—it is the activation window of the capacity to audit. Auditing needs time (slowing down to think), cognitive effort (actively questioning), and awareness of choice (knowing one is choosing). When all friction is removed, all activation conditions of audit vanish simultaneously. Concretely, preserving friction means: no autocompletion, no one-click acceptance, no concealment of uncertainty—retaining at the interface layer the minimal conditions under which users realize they are choosing.

But interface-layer friction is insufficient for institutional-scale problems. §5 has already given the two-sided basis of this conclusion: individual confirmation will be corroded by the automation paradox (Bainbridge), and voluntary self-restraint will be hollowed out by commercial incentives (Marcus, 2024). The direction of treatment must therefore be intervention at both the interface and the institutional layers: preserve friction at

the interface layer, establish independent oversight at the institutional layer.

The Marquand House Collective (2026), in Auditing AI, advances the operationalization of independent oversight one step. Three core principles. First, the gold standard of auditing is third-party independence—no co-authorship, employment, or funding dependence between the auditor and the developer or operator of the audited system—because "this type of independence is necessary to preserve the potential for AI audits to lead to genuine accountability" (Marquand House, 2026). Second, independent auditing needs legal protection. Sandvig v. Barr (2020) was the first case to argue that computer-crime laws could not be applied to independent audits by researchers and journalists (Marquand House, 2026)—the court held that the Computer Fraud and Abuse Act (CFAA) does not criminalize violations of terms of service alone. The U.S. Supreme Court subsequently provided greater clarity in Van Buren v. United States (2021), specifically citing Sandvig and noting the importance of AI audits—what the Court called "online civil rights testing" (Marquand House, 2026, ch. 6). Third, abandonment is a legitimate conclusion of an audit. When Twitter's internal audit found that its image-cropping AI was biased against people with darker skin, the company ultimately chose to abandon the system and return cropping decisions to users. When the Dutch nonprofit Algorithm Audit reviewed the City of Amsterdam's Smart Check welfare-fraud detection system, officials concluded they could not definitively rule out discrimination and ultimately abandoned the system. "While abandoning a system might seem like a failure, it can represent a positive decision" (Marquand House, 2026). Abandonment is not the failure of an audit—it is one of its legitimate conclusions.

The complete governance chain is therefore: preserve friction at the interface layer → establish independent oversight at the institutional layer → give independent audits legal protection → let audit conclusions include abandonment. This chain is no panacea—it is the minimal set of conditions, under the constraints of the current architecture, that makes the closed loop's self-reinforcement more difficult.

One open question remains. The subject of responsibility in M2M settings has begun to collapse: when the user's agent and the merchant's agent conclude a disadvantageous trade, who bears responsibility? Finance has clear legal subjects and quantifiable losses, but harm caused by AI agents to users in everyday settings is often hard to quantify precisely—responsibility diffuses and dissolves into emergence. Current law is not without experience of autonomous interaction—the U.S. E-SIGN Act has provided since 2000 that a contract is not denied legal effect merely because its formation involved the action of one or more electronic agents, so long as that action is legally attributable to the person to be bound—but when damage is multidimensional, hard to price, and the interacting parties are both autonomous systems, how responsibility is allocated, how evidence is obtained, and whether the injured can obtain effective remedy remained open questions in 2026. The Air Canada precedent suggests that agentic commerce will most likely replay the financial path—mandatory kill switches, audit logs, ex-post recovery—rather than the social-media path of immunity. But whether that path extends to multidimensional, hard-to-price everyday settings is a question that in 2026 still has no answer.

The chapter closes with von Neumann. Writing in 1955 in "Can We Survive Technology?," he observed that the human species has been subjected to similar tests before, and that the qualities required can only be specified as "patience, flexibility, intelligence." Just over seventy years on, the patience, flexibility, and intelligence he invoked are precisely the modes of cognition being systematically marginalized by the current environment of choice: deep reading, sustained attention, deferred gratification, cross-source verification—these are the cognitive preconditions of audit, and they are being steadily eroded by the zero-friction environment. What is being marginalized is not merely certain kinds of content—it is certain modes of cognition themselves.

The regression of subjectivity requires no violence—only zero friction.

In this book's language: the zero-friction environment is turning every room into a room with nothing but a clock in it—and auditing is the few steps toward the window.

This chapter concludes the user-side trilogy: "Collecting You → Understanding You → Replacing You"—the full chain, from data collection through behavioral prediction to the outsourcing of judgment and action, has been traced. The zero-friction environment is their common base condition; meta-audit capacity is the user's most direct interface of resistance; and preserving friction and independent oversight are the minimal institutional conditions for keeping that interface from being sealed.

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